University of St.Gallen

Using Computer Simulation

for Understanding Organizations as Complex Systems

Week 1: Introduction

Stephan Aier
Marcel Cahenzli

From insight to impact.

Agenda

Agenda

Stephan

Today

- Professor and senior lecturer at HSG School of Computer Science from 2021, Director at ICV-HSG from 2023
- Co-head Data Management and Analytics Community (and its predecessors) at IWI-HSG from 2006

Past

- Dipl.-Ing. Industrial Engineering and Management, Technical University Berlin in 2002
- Dr.-Ing. Information Systems, Technical University Berlin in 2006
- Post-doc at IWI-HSG in 2006, appointed assistant professor in 2010, *Privatdozent* in 2017, affiliate professor in 2020
- Executive director IWI-HSG 2020-2023

All the time

- Fundamental research in the area of IS architecture, data management and analytics, and digital platforms funded by public research organizations
- Applied research funded by industry partners like ABB, AXA, BOSCH, Credit Suisse, Deutsche Telekom, HP, IBM, Munich Re, Motorola, Novartis, SAP, Swiss Post, Swiss Re, UBS, and others.

… married, two daughters, three turntables, ~~two~~ ~~tree~~ four mountain bikes (and a road bike … and an indoor bike), many hundred vinyl records and even more CDs (although they continue life on some zfs file systems for some time now), likes to engineer and build stuff for relaxing

Marcel

Today

- Executive Director of the Bachelor in Computer Science and the Ph.D. in Computer Science
- Lecturer and Tutor at HSG

Past

- Education: BBWL-HSG (2016), MBI-HSG (2018), Dr. oec HSG (2022)
- Professional: Startup (2013-2015), Marketing- & Strategy-Consultant (2014-2015), Business Development (2016), High-School Teacher replacement
- For fun: oikos project head, TRI Club Bodensee treasurer, DocNet vice president, Familiengartenverein auditor

All the time

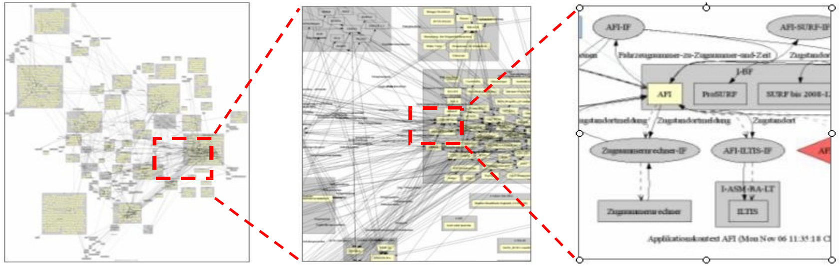
- Sports, (board) games, being outdoors, camping
- Crafts, buidling things (3D printer, coding, garden shed, …)

Agenda

Using Computer Simulation for Understanding Organizations as Complex Systems

What is (a) complex (system)?

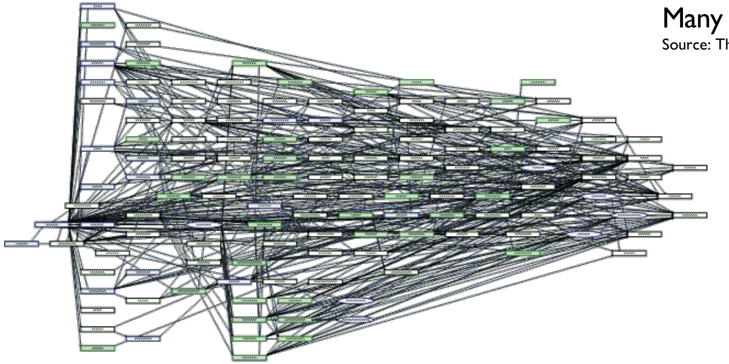
What properties does a complex system have?



Many Objects Source: Rytz, Bernhard: Angewandte Architekturplanung, SBB, AWF 2010

Many structural dependencies
Source: The Atlas of Economic Complexity

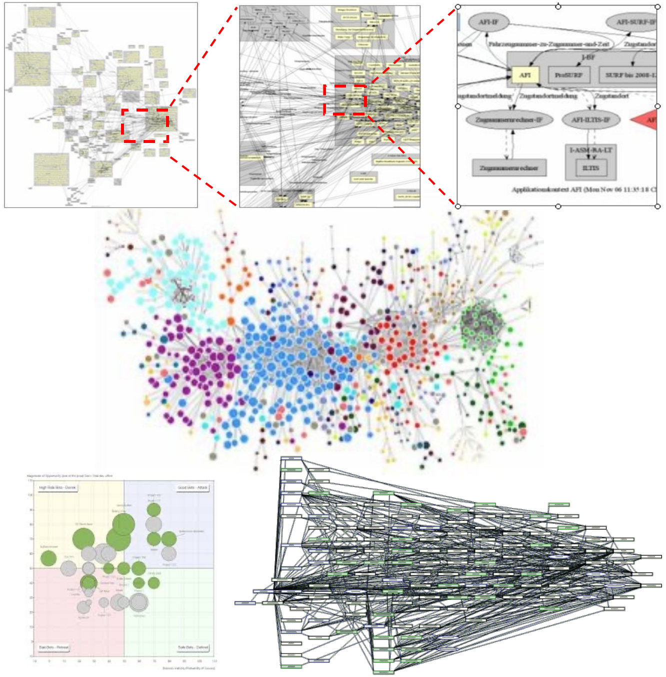
Many changes



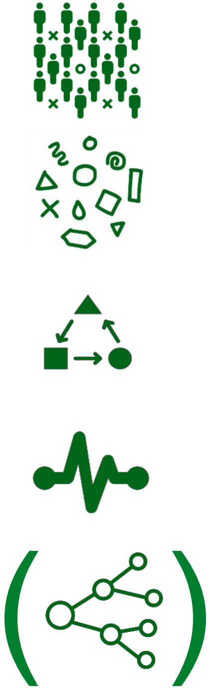
Many dynamic dependencies

What else characterizes a complex (adaptive) system?

Complexity of socio-technical systems



Characteristics of complex adaptive systems (CAS)



- Large number of elements
- Diversity
- Interdependences
- Non-linearity
- Emergence

Axelrod, R., & Cohen, M. D. (1999). Harnessing Complexity: Organizational Implications of a Scientific Frontier. Free Press.

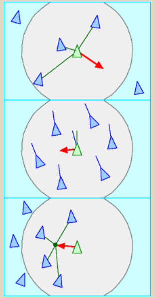
For an alternative but compatible summary see: https://complexityexplained.github.io/

Flock of birds

Source: National Geographic, https://youtu.be/V4f_1_r80RY

How to describe the behavior of the system?

- there is no central control
- each bird behaves autonomously and "follows" some simple rules:
  - **Separation:** avoid crowding neighbours (short range repulsion)
  - **Alignment:** steer towards average heading of neighbours
  - **Cohesion:** steer towards average position of neighbours (long range attraction)

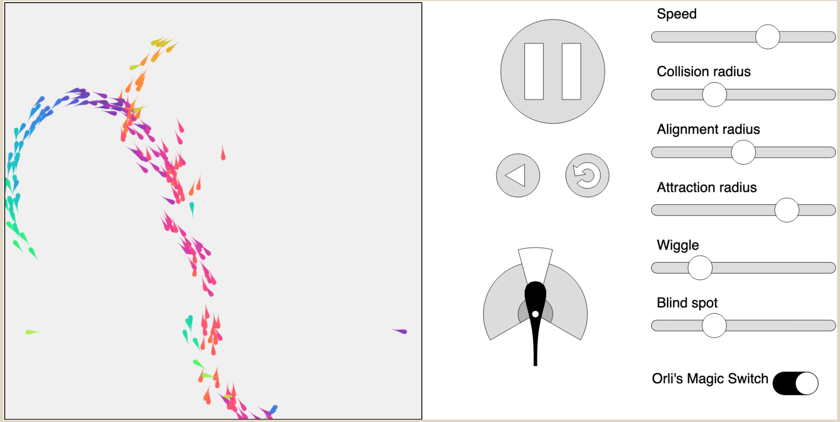


Source: https://en.wikipedia.org/wiki/Flocking, https://en.wikipedia.org/wiki/Boids,

Craig W. Reynolds. 1987. Flocks, herds and schools: A distributed behavioral model. In Proceedings of the 14th annual conference on Computer graphics and interactive techniques (SIGGRAPH '87). Association for Computing Machinery, New York, NY, USA, 25–34. https://doi.org/10.1145/37401.37406

More general collective behavior and swarming models

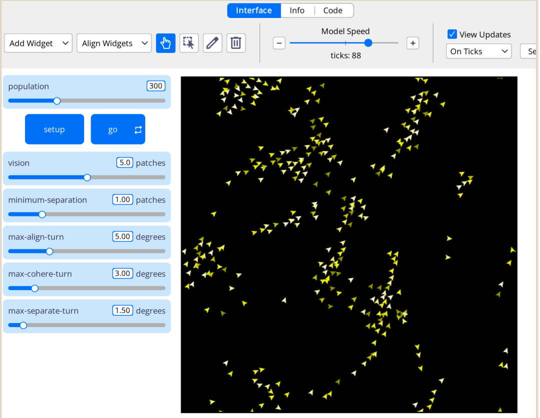
Complexity Explorables



Source: https://www.complexity-explorables.org/explorables/flockn-roll/

→ This is nice to play with and explore diverse models of complex systems.

Netlogo




```
turtles-own [
  flockmates         ;; agentset of nearby turtles
  nearest-neighbor   ;; closest one of our flockmates
]

to setup
  clear-all
  create-turtles population
    [ set color yellow - 2 + random 7  ;; random shades look nice
      set size 1.5  ;; easier to see
      setxy random-xcor random-ycor
      set flockmates no-turtles ]
  reset-ticks
end

to go
  ask turtles [ flock ]
  ;; the following line is used to make the turtles
  ;; animate more smoothly.
  repeat 5 [ ask turtles [ fd 0.2 ] display ]
  ;; for greater efficiency, at the expense of smooth
  ;; animation, substitute the following line instead:
  ;;   ask turtles [ fd 1 ]
  tick
end

to flock  ;; turtle procedure
  find-flockmates
  if any? flockmates
    [ find-nearest-neighbor
      ifelse distance nearest-neighbor < minimum-separation
        [ separate ]
        [ align
          cohere ] ]
end

to find-flockmates  ;; turtle procedure
  set flockmates other turtles in-radius vision
end

to find-nearest-neighbor ;; turtle procedure
  set nearest-neighbor min-one-of flockmates [distance myself]
end

;;; SEPARATE

to separate  ;; turtle procedure
  turn-away ([heading] of nearest-neighbor) max-separate-turn
end

;;; ALIGN

to align  ;; turtle procedure
  turn-towards average-flockmate-heading max-align-turn
end

to-report average-flockmate-heading  ;; turtle procedure
  ;; We can't just average the heading variables here.
  ;; For example, the average of 1 and 359 should be 0,
  ;; not 180.  So we have to use trigonometry.
  let x-component sum [dx] of flockmates
  let y-component sum [dy] of flockmates
  ifelse x-component = 0 and y-component = 0
    [ report heading ]
    [ report atan x-component y-component ]
end

;;; COHERE

to cohere  ;; turtle procedure
  turn-towards average-heading-towards-flockmates max-cohere-turn
end

to-report average-heading-towards-flockmates  ;; turtle procedure
  ;; "towards myself" gives us the heading from the other turtle
  ;; to me, but we want the heading from me to the other turtle,
  ;; so we add 180
  let x-component mean [sin (towards myself + 180)] of flockmates
  let y-component mean [cos (towards myself + 180)] of flockmates
  ifelse x-component = 0 and y-component = 0
    [ report heading ]
    [ report atan x-component y-component ]
end
;;; HELPER PROCEDURES

to turn-towards [new-heading max-turn]  ;; turtle procedure
  turn-at-most (subtract-headings new-heading heading) max-turn
end

to turn-away [new-heading max-turn]  ;; turtle procedure
  turn-at-most (subtract-headings heading new-heading) max-turn
end

;; turn right by "turn" degrees (or left if "turn" is negative),
;; but never turn more than "max-turn" degrees
to turn-at-most [turn max-turn]  ;; turtle procedure
  ifelse abs turn > max-turn
    [ ifelse turn > 0
        [ rt max-turn ]
        [ lt max-turn ] ]
    [ rt turn ]
end


; Copyright 1998 Uri Wilensky.
; See Info tab for full copyright and license.
```

Source: Wilensky, U. (1998). NetLogo Flocking model. http://ccl.northwestern.edu/netlogo/models/Flocking. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

→ This is nice to create and explore models of complex systems.

“Complex systems theory (…) defines complex systems as systems that are composed of multiple individual elements that interact with each other yet whose aggregate properties or behaviour is not predictable from the elements themselves.”

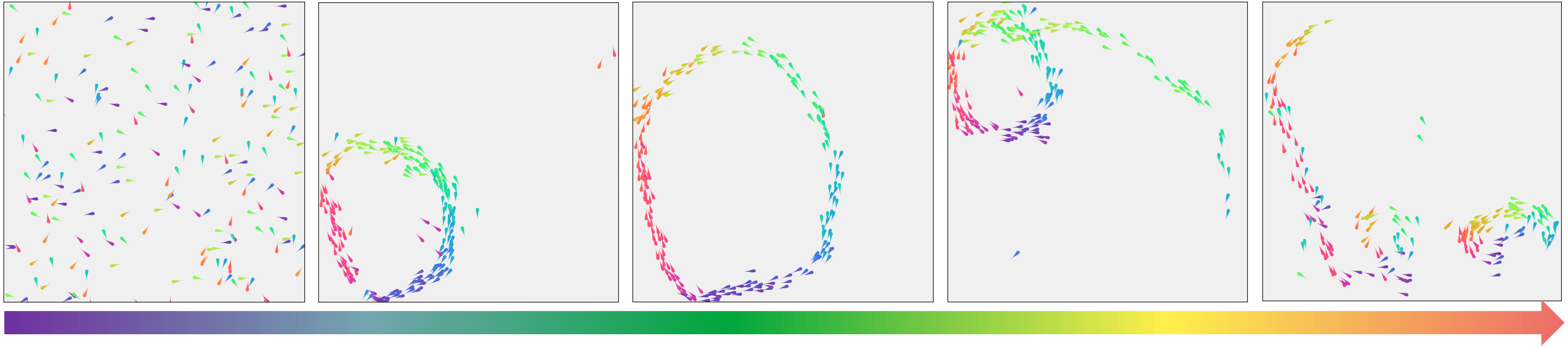
(Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press. p.6)

What does this mean: “aggregate … behaviour is not predictable from the elements themselves”?

Emergence and self-organization

Emergence: *“the arising of novel and coherent structures, patterns, and properties through the interactions of multiple distributed elements”* (Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press. p.6)

https://www.complexity-explorables.org/explorables/flockn-roll/

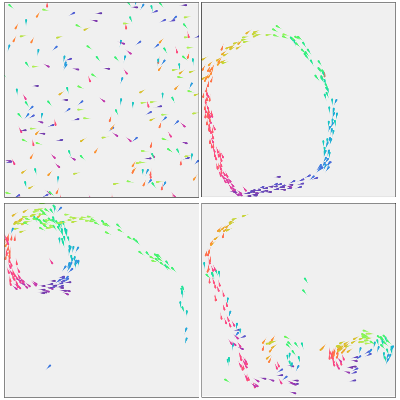


Self-organization: “*In complex systems, order can emerge without any design or designer*” (Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press. p.7)

Macro-level patterns stem from rules at the micro level. Formation processes include some randomness that helps create order.

Complex Adaptive Systems (CAS)

Complexity, emergence and self-organization



Traditional approaches (e.g. more detailed analysis of subsystems, long-term statistical behavior, fixed interaction models, …) fail to capture system behavior or are "difficult" to formalize.

Newer theories (CAS) focus on:

- Communication and interaction patterns
- Non-rigid structures
- Local behavior
- Global properties and pervasive attributes (e.g. culture, system behavior)

Goal:
Analyze, adapt and apply successful approaches from other fields to our problems

More examples from diverse fields?

The Prisoner's Kaleidoscope
The prisoner's dilemma game on a lattice

Spin Wheels
Phase-coupled oscillators on a lattice

Eigenartig
The spatial hypercycle model

Come Together
Chemotaxis in Dictyostelium discoideum

Thrilling Milling Schelling Herings
Swarming behavior of animals that seek their kin

T. Schelling plays Go
The Schelling model

Nah dah dah nah nah... (Opus, 1984)
Conway's Game of Life

Jujujajáki networks
The emergence of communities in weighted networks

Horde of the Flies
The Vicsek-Model

The walking head
Pedestrian dynamics

Swårmalätørs
Oscillators that sync and swarm

Clustershuck
A Network Growth Model that Naturally Yields Clustered Networks

Repliselmut
Yet another Complexity Explorable on evolution

Janus Bunch
Dynamics of two-phase coupled oscillators

Berlin 8:00 a.m.
The emergence of phantom traffic jams

Prime Time
The distribution of primes along number spirals

I sing well-tempered
The Ising Model

Hopfed Turingles
Pattern Formation in a simple reaction-diffusion system

Critically Inflammatory
A forrest fire model

Facebooked Flu Shots
Network vaccination

Stranger Things
Strange attractors

Scott's World*
Microbial growth patterns

Echo Chambers
A model for opinion dynamics

Surfing a Gene Pool
Expansion of clones with idential fitness

Anomalous Itinerary
Lévy flights

Albert & Carl Friedrich
Random Walks & Diffusion

Dr. Fibryll & Mr. Glyde
Pulse-coupled oscillators

The Blob
A network's giant component

I herd you!
How herd immunity works

Yo, Kohonen!
Kohonen's Self-Organizing Map

Kelp!!!
A stochastic cellular automaton

Knitworks
Growing complex networks

If you ask your XY
The XY model of statistical mechanics

Maggots in the Wiggle Room
The dynamics of evolution

Into the Dark
Collective intelligence

Lotka Martini
The Lotka-Volterra model

Critical HexSIRSize
The stochastic, spatial SIRS model

Hokus Fractus!
Famous fractals

Cycledelic
The spatial rock-paper-scissors game

Epidemonic
The SIRS epidemic model

Weeds & Trees
Lindenmayer Systems

Ride my Kuramotocycle!
The Kuramoto model

A Patchwork Darwinge
Evolution: Variation and Selection

Barista's Secret
Percolation on a lattice

Double Trouble
The double pendulum

Flock'n Roll
Collective behavior and swarming

Particularly Stuck
Diffusion Limited Aggregation

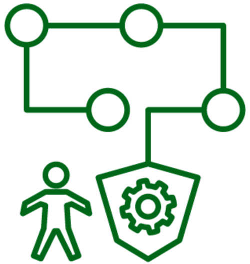
Kick it like Chirikov
The kicked rotator (standard map)

Keith Haring's Mexican Hat
Pattern Formation by Local Excitation and Long-Range Inhibition

https://www.complexity-explorables.org/

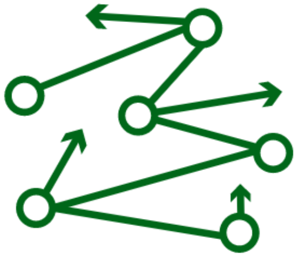
Using Computer Simulation for Understanding Organizations as Complex Systems

How can complex organizations be “guided”?



Guiding

as a balance between



Top-down control, or: designed by a designer

Bottom-up emergence, or: self-organized

Enforced through

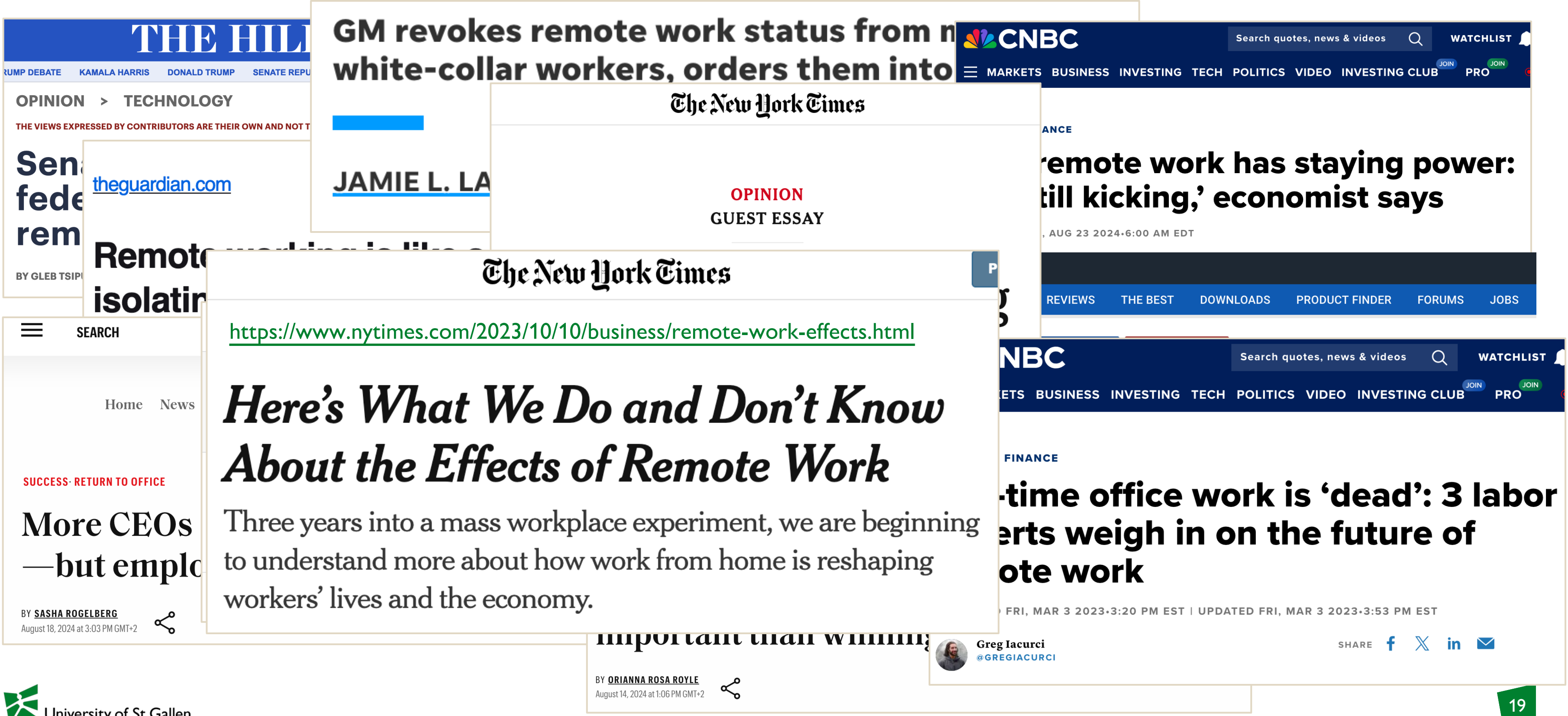
- Rules
- Principles
- Project reviews
- …
- Sanctions

Driven by

- Individuals’ goals and behavior
- Expectations of peers
- Successful practices
- Norms and values of the organization
- …

Haki, K., Beese, J., Aier, S., & Winter, R. (2020). The Evolution of Information Systems Architecture: An Agent-Based Simulation Model. MIS Quarterly, 44(1), 155-184.

To work remotely or from the office?



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theguardian.com
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isolatin

GM revokes remote work status from m white-collar workers, orders them into
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The New York Times
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remote work has staying power: till kicking,' economist says
, AUG 23 2024·6:00 AM EDT
REVIEWS THE BEST DOWNLOADS PRODUCT FINDER FORUMS JOBS

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SUCCESS · RETURN TO OFFICE
More CEOs —but emplo
BY SASHA ROGELBERG
August 18, 2024 at 3:03 PM GMT+2

The New York Times
https://www.nytimes.com/2023/10/10/business/remote-work-effects.html
Here's What We Do and Don't Know About the Effects of Remote Work
Three years into a mass workplace experiment, we are beginning to understand more about how work from home is reshaping workers' lives and the economy.

important than winnin
BY ORIANNA ROSA ROYLE
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-time office work is 'dead': 3 labor erts weigh in on the future of ote work
FRI, MAR 3 2023·3:20 PM EST | UPDATED FRI, MAR 3 2023·3:53 PM EST
Greg Iacurci
@GREGIACURCI
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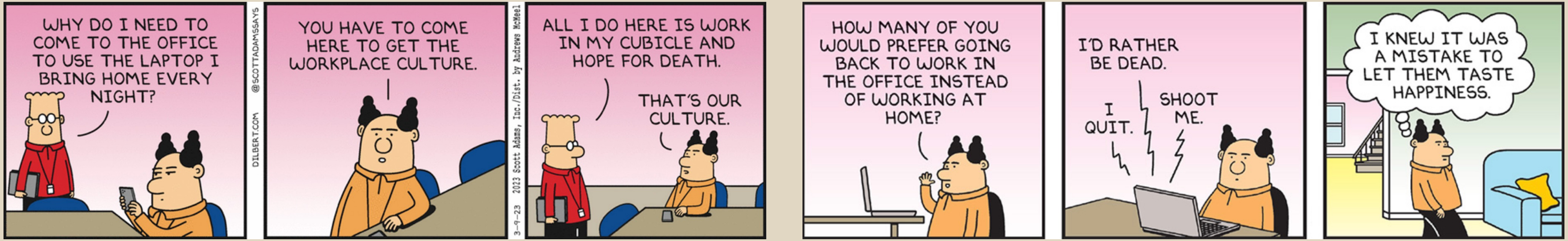
Short Assignment

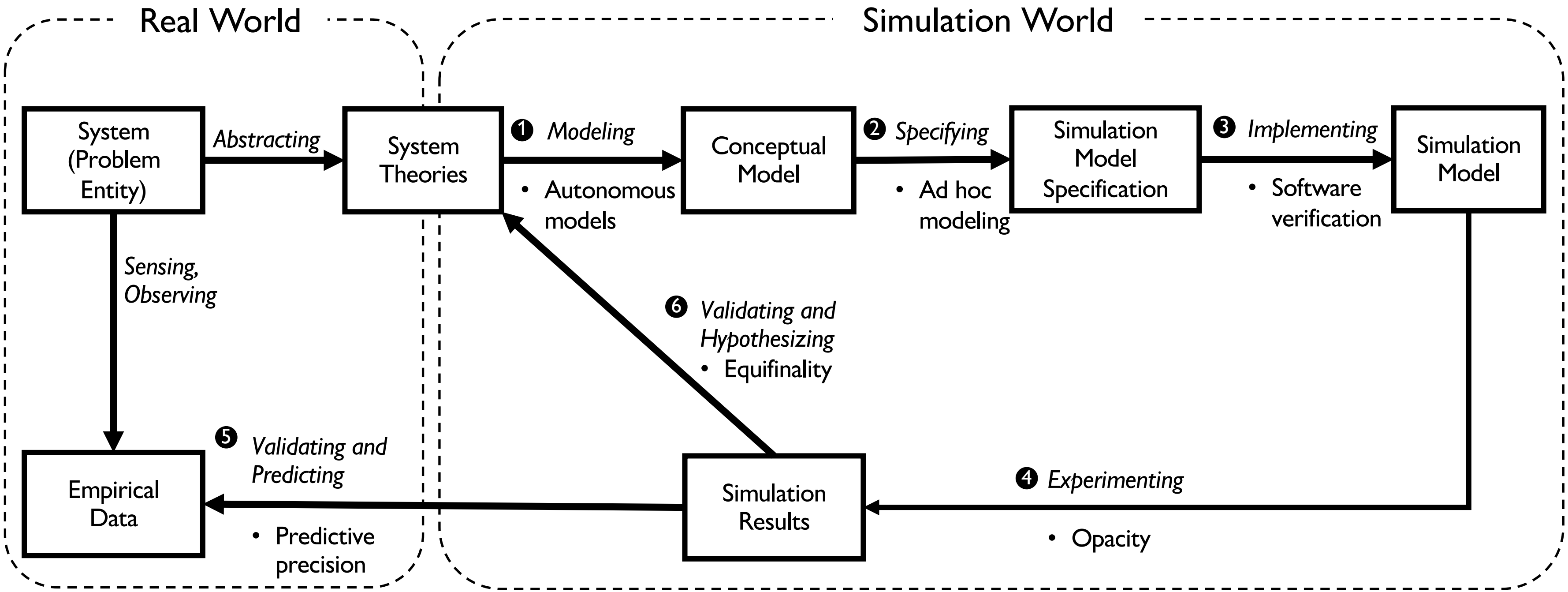
1. Discuss in small groups with your neighbor(s):
   How would the "home office approach" develop in a company and what would be the effects? Breakdown the question into sub-questions:
   a. What are design options for working from home?
   b. What are the goals of stakeholder groups (such as managers, employees, …)
   c. What are possible/expected outcomes of your design options, and do they meet the stakeholders' goals?
   d. (What cause-effect relations do assume and how would you operationalize those into means-ends relations?)

2. How can you more systematically research the phenomenon?



Using Computer Simulation for Understanding Organizations as Complex Systems

A process model of simulation-based research



Beese, J., Haki, M. K., Aier, S., & Winter, R. (2019). Simulation-Based Research in Information Systems: Epistemic Implications and a Review of the Status Quo. Business & Information Systems Engineering, 61(4), 503-521.

Agenda

Structure

Date	Topic	Deliverables	Literature	Responsible
19.02.	Introduction to the course, theory foundations, simulation demos	-	papers 1, 2	Stephan & Marcel
26.02.	How to apply agent-based modelling?	-	book chapter 1	Marcel
05.03.	How to implement simulation models with NetLogo?	-	book chapter 2 (3, 4, 5)	Marcel
12.03.	Independent case work (*no input session*)	-		-
19.03.	Intermediate group presentation and feedback on case/model	Presentation		Stephan & Marcel
26.03.	How to run simulation experiments and analyse simulation data?	-	book chapter 6	Stephan
	(*break*)			
16.04.	Intermediate group presentation on implemented model and analysis	Presentation		Stephan & Marcel
23.04.	How to verify and validate simulation models?	-	book chapter 7	Stephan
30.04.	Independent case work (*no input session*)	-		-
07.05.	Final case presentation in groups and reflection of learnings	Presentation, Model, Analysis		Stephan & Marcel
14.05.	Ascension Day (*no input session*)	-		-
21.05.	Oral exams	Oral Exam		Stephan & Marcel

Paper 1: Beese, J., Haki, M. K., Aier, S., & Winter, R. (2019). Simulation-Based Research in Information Systems: Epistemic Implications and a Review of the Status Quo. Business & Information Systems Engineering, 61(4), 503-521.
Paper 2: Haki, K., Beese, J., Aier, S., & Winter, R. (2020). The Evolution of Information Systems Architecture: An Agent-Based Simulation Model. MIS Quarterly, 44(1), 155-184.
Book: Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press.

Deliverables

Practically applied:

- 50% Final Presentation (group work, group grade)
- 20% Programming (group work, group grade)

Conceptual understanding:

- 30% Oral Exam (group work, **individual** grade)

University of St.Gallen

Course and Examination Fact Sheet: Spring Semester 2026

8,727: Technologien/Technologies: Using Computer Simulation for Understanding Organizations as Complex Systems

ECTS credits: 6

Overview examination/s

(binding regulations see below)
decentral - Programming, Digital, Group work group grade (20%)
Examination time: Term time
decentral - Presentation, Analog, Group work group grade (50%)
Examination time: Term time
decentral - Oral examination and technical discussions, Digital, Group work individual grade (30%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
8,727,1.00 Technologien/Technologies: Using Computer Simulation for Understanding Organizations as Complex Systems -- English -- Cahenzli Marcel , Aier Stephan

About the Final Presentation (50%)

- 15 min presentation + Q&A Session
- Group size: 2-4
- Content of the presentation:
  - Clearly described phenomenon (incl. setting, stakeholders, their goals, courses of action, research questions)
  - Conceptual model (incl. agents, environment, interaction behaviour; incl. grounding and justification)
  - Independent and dependent variables
  - Pseudo Code
  - Live Demo
  - Model Validation
  - Results (addressing the RQ, insights from your data)
  - Discussion (implications for practice and research, limitations, key take-aways)

About the Programming Part (20%)

What to submit?

- NetLogo File or equivalent
  - interface (*Interface* tab)
  - explanatory description (*Info* tab)
  - documented code (*Code* tab)
  - experiment definition (Tools → *BehaviorSpace*)
- documented scripts for data analysis (e.g., in Python, Excel)
- data set (data generated through simulation experiments) on which the analysis (presentation) is based

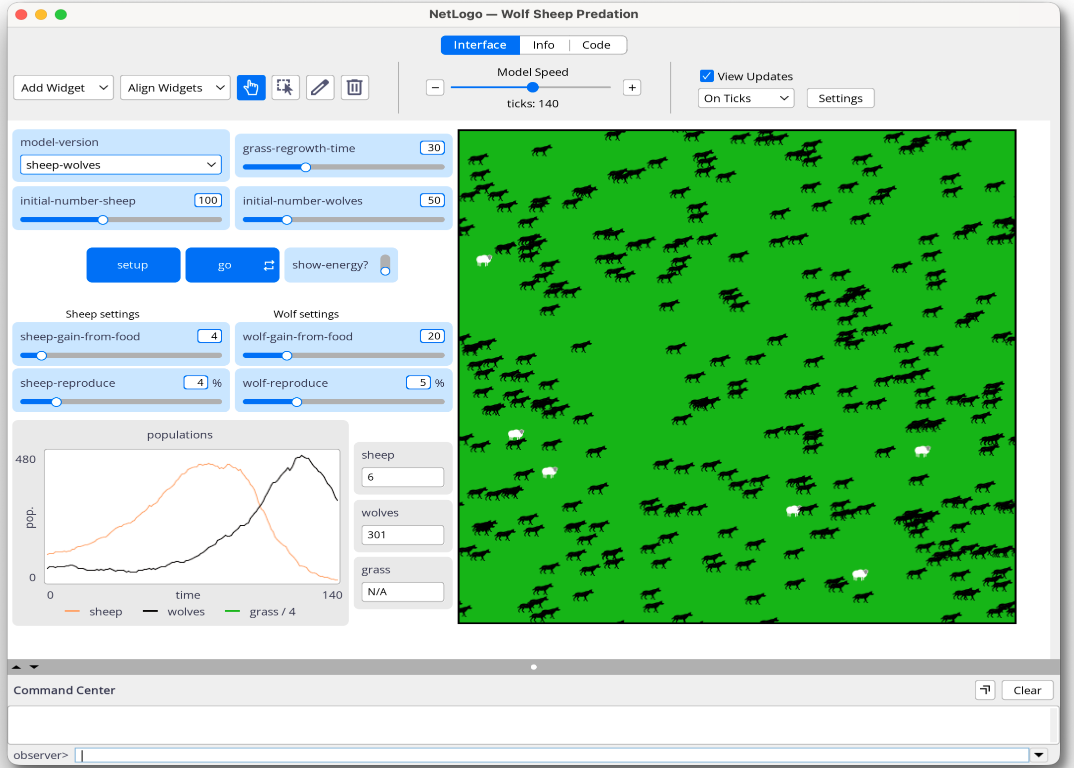
Criteria

- needs to run without errors
- understandable as a standalone documentation

PS: If any significant changes are made after the presentation, a change summary needs to be added to the info tab.

NetLogo

Interface tab



Info tab

NetLogo — Wolf Sheep Predation

Interface | Info | Code

Find | Edit

WHAT IS IT?

This model explores the stability of predator-prey ecosystems. Such a system is called unstable if it tends to result in extinction for one or more species involved. In contrast, a system is stable if it tends to maintain itself over time, despite fluctuations in population sizes.

HOW IT WORKS

There are two main variations to this model.

In the first variation, the “sheep-wolves” version, wolves and sheep wander randomly around the landscape, while the wolves look for sheep to prey on. Each step costs the wolves energy, and they must eat sheep in order to replenish their energy - when they run out of energy they die. To allow the population to continue, each wolf or sheep has a fixed probability of reproducing at each time step. In this variation, we model the grass as “infinite” so that sheep always have enough to eat, and we don’t explicitly model the eating or growing of grass. As such, sheep don’t either gain or lose energy by eating or moving. This variation produces interesting population dynamics, but is ultimately unstable. This variation of the model is particularly well-suited to interacting species in a rich nutrient environment, such as two strains of bacteria in a petri dish (Gause, 1934).

The second variation, the “sheep-wolves-grass” version explicitly models grass (green) in addition to wolves and sheep. The behavior of the wolves is identical to the first variation, however this time the sheep must eat grass in order to maintain their energy - when they run out of energy they die. Once grass is eaten it will only regrow after a fixed amount of time. This variation is more complex than the first, but it is generally stable. It is a closer match to the classic Lotka Volterra population oscillation models. The classic LV models though assume the populations can take on real values, but in small populations these models underestimate extinctions and agent-based models such as the ones here, provide more realistic results. (See Wilensky & Rand, 2015; chapter 4).

The construction of this model is described in two papers by Wilensky & Reisman (1998; 2006) referenced below.

HOW TO USE IT

1. Set the model-version chooser to “sheep-wolves-grass” to include grass eating and growth

NetLogo

Code tab

NetLogo — Wolf Sheep Predation

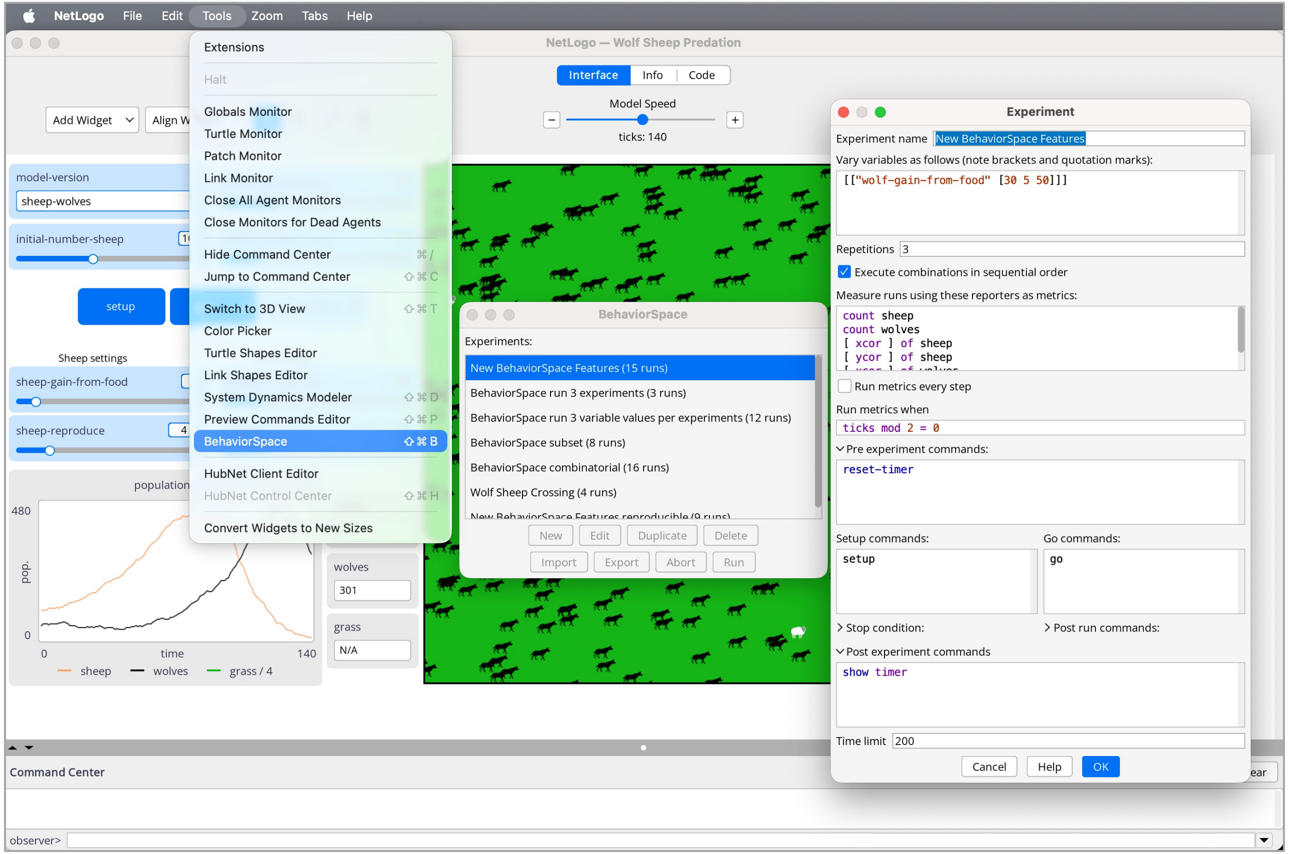
Interface | Info | Code

Check | Find | Procedures | Included Files | Separate code window | Preferences

```
globals [ max-sheep ]  ; don't let the sheep population grow too large

; Sheep and wolves are both breeds of turtles
breed [ sheep a-sheep ]  ; sheep is its own plural, so we use "a-sheep" as the singular
breed [ wolves wolf ]

turtles-own [ energy ]       ; both wolves and sheep have energy

patches-own [ countdown ]    ; this is for the sheep-wolves-grass model version

to setup
  clear-all
  ifelse netlogo-web? [ set max-sheep 10000 ] [ set max-sheep 30000 ]

  ; Check model-version switch
  ; if we're not modeling grass, then the sheep don't need to eat to survive
  ; otherwise each grass' state of growth and growing logic need to be set up
  ifelse model-version = "sheep-wolves-grass" [
    ask patches [
      set pcolor one-of [ green brown ]
      ifelse pcolor = green
        [ set countdown grass-regrowth-time ]
      [ set countdown random grass-regrowth-time ] ; initialize grass regrowth clocks randomly for brown patches
    ]
  ]
  [
    ask patches [ set pcolor green ]
  ]

  create-sheep initial-number-sheep  ; create the sheep, then initialize their variables
  [
    set shape  "sheep"
    set color white
    set size 1.5  ; easier to see
    set label-color blue - 2
    set energy random (2 * sheep-gain-from-food)
    setxy random-xcor random-ycor
  ]

  create-wolves initial-number-wolves  ; create the wolves, then initialize their variables
  [
    set shape "wolf"
    set color black
    set size 2  ; easier to see
    set energy random (2 * wolf-gain-from-food)
    setxy random-xcor random-ycor
  ]
  display-labels
  reset-ticks
end
```

NetLogo

BehaviorSpace



About the Oral Examination (30%)

Format: Oral exam, held in groups, online, with individual grades.
Date: Thursday, **21.05.2026**, 14:15-18:00 (regular lecture slot)

Purpose: The exam is geared toward measuring the individual attainment of the learning goals, with a focus on concepts and reflection (as opposed to practical implementation). Questions are therefore directed to individual students.

Content

- All major concepts from the course (what, why, how)
- Reflection on your key learnings from the course (incl. how you can/will transfer insights)
- Questions about the projects

Agenda

Assignment

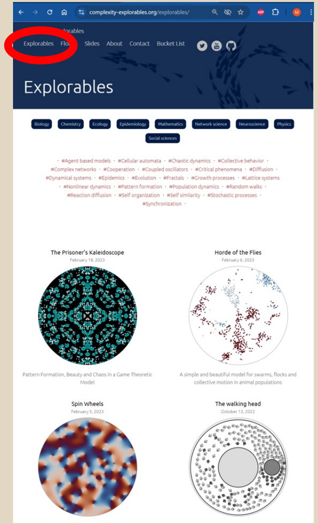
1. Visit complexity-explorables.org: Individually open and play around with models. Choose **two models** that interest you and find out what they do. (15 min)
    1. What is being modeled?
    2. What are the model's components?
    3. What are your configuration options? (i.e., independent variables)
    4. What can you observe? (… can you identify emergent behaviour?)
    5. How could the model behaviour be mapped to an organisation/mgmt?

2. Form groups of four and introduce yourselves. (< 5 min)

3. Identify 4 different models in your group and discuss them based on the questions above. (20 min)

4. Decide on one model you like best as a group, analyse the model's behaviour and prepare a pitch addressing the questions above. (10 min)

5. Present and discuss the model to the class. (2 min / group)



Agenda

Week 2: Applied ABM

Part 1: Theoretical

- Conceptual understanding of Agent-Based-Modelling
  - What is a ‘model’ and what is an ABM specifically?
  - What are the constituents of an ABM?
  - What can (and cannot) be achieved with an ABM?
- Mapping ABM concepts to real-world phenomena
  - Selecting a phenomenon
  - Narrowing down its scope
  - Identifying actors, environments, and behaviours
  - Visualizing the (conceptual) ABM

Week 2: Applied ABM

Part 2: Practical

- Identify 2 organisational phenomena (individually)
- Form groups of 2 and pitch the phenomena to each other
- With these 4 phenomena, pick one and go through the process presented in part 1:
  - Selecting a phenomenon
  - Narrowing down its scope
  - Identifying actors, environments, and behaviours
  - Visualizing the (conceptual) ABM
- Present your work in front of the class.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“From insight to impact”